HamidReza Saadi Dadmarzi

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Education

Koç University, Istanbul, Türkiye Fall 2026–present

Ph.D. in Computational Sciences and Engineering, advised by Assoc. Prof. Mehmet Cengiz Onbaşlı.

Koç University, Istanbul, Türkiye Nov. 2023–Sep. 2026

M.Sc. in Computer Science and Engineering, advised by Prof. Alptekin Küpçü.
Thesis Secure Single Password Authentication Protocols in Distributed Systems.
Defended 28 September 2026. Designed and evaluated UpSPA and its quorum-based extension RoUpSPA.
Selected coursework Modern Cryptography, Elliptic Curves and Cryptography, Communication Networks, Advanced Optimization Methods and Advanced Stochastic Optimization, and Medical Image Analysis.

Amirkabir University of Technology (Tehran Polytechnic) Sep. 2017–Aug. 2020

M.Sc. in Pure Mathematics. Research on quantum error-correcting codes, surface codes, and information-theoretic cryptography. Selected courses included Real Analysis, Advanced Algebra, Information Theory and Algebraic Coding Theory.

Amirkabir University of Technology (Tehran Polytechnic) Sep. 2016–Aug. 2018

Minor in Physics.
Selected courses Electromagnetism I and II, Quantum Mechanics I and II, Mathematical Physics I,II and III, Advanced Mathematical Physics and Advanced Electodynamics I.

Amirkabir University of Technology (Tehran Polytechnic) Sep. 2014–Aug. 2017

B.Sc. in Applied Mathematics. Ranked in the top 5% of the undergraduate cohort.
Selected courses Fundamentals of Matrices and Linear Algebra, Mathematical Analysis, Elementary Theory of Numbers, Fundamentals of Probability, Numerical Analysis, Fundamentals of Algebra, and General Topology.

Selected Publications and Manuscripts

- **Structure Matters: Differentially Private Spectral Statistics for ECG Time Series**.
  Reza Saadi and Erfan Salavati. Accepted poster, TS-LIMITS workshop at NeurIPS 2026.
- **SPA-Web: Browser-Based Single-Password Authentication**.
  Reza Saadi, Ozan Kaymak, Efe Bekteş, and Alptekin Küpçü. Accepted at IEEE ISC Türkiye 2026.
- **Secure and Updatable Single Password Authentication (UpSPA)**.
  Devriş İşler, HamidReza Saadi Dadmarzi, and Alptekin Küpçü. IACR ePrint 2026/784.
  Submitted to ACNS 2027 (Cycle 1).
- **RoUpSPA: Robust Updatable Single Password Authentication**.
  Preprint prepared for ACM Transactions on Privacy and Security (TOPS).

Selected Research

Post-Quantum Cryptography and Quantum Information

- **AEGIS–VAKE.** Structural checks for authenticated key exchange, with Rust and Lean agreeing on all 188 tested executions. Reused a Lean signing-origin proof for two controlled signed Diffie–Hellman revisions under trusted credentials and a symbolic attacker model.
- **Timing Leakage in Post-Quantum Implementations.** Instruction analysis and end-to-end tests of known Kyber timing effects on Intel processors.
- **Computational MUBs in $\mathbb{C}^6$.** Riemannian minimax search for four approximately mutually unbiased bases, with high-precision refinement and rational upper-bound certificates.

Privacy-Preserving Machine Learning and ML Theory

- **PersonGuard and CrossLead.** Ongoing follow-ups to *Structure Matters* on auditable person-level DP for ECG spectra under public-to-private data mismatch and private release of cross-lead structure.
- **Grassmann/PCA Machine Unlearning.** Geometric deletion certificates and efficient updates for randomized PCA with repeated or nearly repeated eigenvalues.

Secure Multiparty Computation and Theoretical Cryptography

- **UpSPA Family.** Threshold authentication with independent credential updates, quorum writes and recovery in RoUpSPA, and browser integration in SPA-Web.

Selected GitHub Projects

- **VAKE.** Rust and Lean toolkit for checking typed protocol models and proof reuse. GitHub
- **Structure Matters.** Python code and results for differentially private ECG spectral release. GitHub
- **UpSPA-Web.** Chrome browser benchmarks with Rust/WASM, Go, and PostgreSQL. GitHub
- **Threshold Password Authentication Protocols.** Rust network benchmarks for four protocols. GitHub

Research and Professional Experience

Koç University, Istanbul, Türkiye 2023–2026 (M.Sc. research)

Research Assistant, KU-Crypto Lab. Designed, analyzed, implemented, and benchmarked threshold authentication protocols with Prof. Alptekin Küpçü.

CURE implementation contribution. Supported Go/Lattigo implementation and debugging for homomorphic-encryption-based split learning, including parallel encrypted experiments. Paper

BTL Coin Crypto Exchange, Istanbul / remote May 2021–Oct. 2023

Software Developer. Built personalized market screeners and automated crypto-asset management tools using Go and Docker.

Institute for Research in Fundamental Sciences (IPM), Tehran Oct. 2020–Mar. 2021

Quantum machine learning research on quantum kernel optimization for neural networks with Qiskit and PyQuil.

Paid University Project, Tehran May 2019–Jul. 2020

Developed a data warehouse and integrated analytics workflows using Excel and Tableau.

Teaching Experience

Koç University Teaching Assistant

Modern Cryptography	Fall 2026, Fall 2025
Computer Networks & Security	Spring 2026, Spring 2025
Data Structures & Algorithms	Fall 2024
Discrete Structures	Spring 2024
Introduction to Python Programming	Fall 2023

Amirkabir University of Technology Teaching Assistant

Time Series Analysis	Fall 2019
Multivariable Calculus	Spring 2019
Introduction to C Programming	Fall 2018
Partial Differential Equations	Fall 2018
Foundation of Matrix and Linear Algebra	Spring 2018
One Variable Calculus	Fall 2017
Introduction to Probability	Spring 2017
General Physics (Electricity and Magnetism)	Spring 2017
Foundation of Mathematical Analysis	Spring 2017
Introduction to Set Theory and Logic	Fall 2016–Spring 2018

Mentoring and Academic Service

- Mentored five interns across multiple semesters.
- Reviewed for IEEE TIFS (since 2024), ACM ASIACCS (2024, 2026), ESORICS (2025, 2026), ACNS (2026), Royal Society Open Science (2024), PeerJ (2026), JISA (2025), and Turkish Journal of Electrical Engineering & Computer Sciences (2025).

Honors and Awards

- Outstanding Teaching Assistant Award, Koç University (2025).
- Exceptional Talent Center recognition for double-major eligibility and admission to graduate study without the national entrance examination.

Technical Skills and Languages

Programming and verification Rust, Python, Go, MATLAB, and Lean 4.
Tools and libraries Docker, LaTeX, Qiskit, PyQuil, Lattigo, SciPy, and CVXPY.
Languages Persian (native), English (working professional), and Turkish (elementary).

Conferences and Workshops

- 5th and 6th Frontiers in Mathematical Sciences Conferences, IPM, Tehran (July 2017 and July 2018).
- Workshop on Mathematics of Information-Theoretic Cryptography, IMS, Singapore (September 2016).

References

Prof. Alptekin Küpçü (email) | Dr. Erfan Salavati (email) | Assoc. Prof. Mehmet Cengiz Onbaşlı (email).